Sample Answers

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Note: Problems marked with an asterisk (*) are optional and are less likely to be tested in the exams. You can safely skip them if you don't want to spend much time on the worksheet.

1. Inverse Functions

***1.** Given the function $f(x) = x^3 + 6x^2 + 12x + 5$, find its inverse function $f^{-1}(x)$.

Solution:

We begin by setting $y = f(x)$. Our goal is to rearrange the expression to solve for x in terms of y.

$$y = x^3 + 6x^2 + 12x + 5$$

The expression on the right resembles the expansion of a binomial cube, $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$. We can rewrite $f(x)$ by completing the cube:

$$\begin{aligned} y &= (x^3 + 6x^2 + 12x + 8) - 3 \\ y &= (x+2)^3 - 3 \end{aligned}$$

Now, we can solve for x:

$$\begin{aligned} y + 3 &= (x+2)^3 \\ \sqrt[3]{y+3} &= x + 2 \\ x &= \sqrt[3]{y+3} - 2 \end{aligned}$$

To find the inverse function $f^{-1}(x)$, we swap the variables x and y:

$$f^{-1}(x) = \sqrt[3]{x+3} - 2$$

***2.** Given the function $f(x) = \frac{x}{\sqrt{1-x^2}}$ with a domain of $(-1, 1)$, find its inverse function $f^{-1}(x)$.

Solution:

Let $y = f(x)$. To find the inverse, we swap x and y and then solve for the new y.

$$x = \frac{y}{\sqrt{1-y^2}}$$

To eliminate the square root, we square both sides of the equation:

$$x^2 = \frac{y^2}{1-y^2}$$

Now, we rearrange the equation to isolate y^2:

$$\begin{aligned}
x^2(1-y^2) &= y^2 \\
x^2 - x^2y^2 &= y^2 \\
x^2 &= y^2 + x^2y^2 \\
x^2 &= y^2(1+x^2) \\
y^2 &= \frac{x^2}{1+x^2}
\end{aligned}$$

Taking the square root of both sides gives two possible solutions for y:

$$y = \pm\sqrt{\frac{x^2}{1+x^2}} = \pm\frac{|x|}{\sqrt{1+x^2}}$$

To determine the correct sign, we observe the original function $f(x) = \frac{x}{\sqrt{1-x^2}}$. The denominator is always positive, so the sign of $f(x)$ is the same as the sign of x. The inverse function $f^{-1}(x)$ must also have this property. Therefore, we must choose the sign such that y has the same sign as x. This is achieved by taking the positive root and replacing $|x|$ with x.

$$f^{-1}(x) = \frac{x}{\sqrt{1+x^2}}$$

***3.** If $f(x) = \sqrt[3]{x+\sqrt{x^2+1}} + \sqrt[3]{x-\sqrt{x^2+1}}$, find $f^{-1}(x)$.

Hint: $a^3+b^3 = (a+b)(a^2-ab+b^2)$ *and* $(a+b)^3 = a^3+b^3+3ab(a+b)$

Solution:

Let $y = f(x)$. We can simplify the expression by letting $a = \sqrt[3]{x+\sqrt{x^2+1}}$ and $b = \sqrt[3]{x-\sqrt{x^2+1}}$, so $y = a+b$. To eliminate the cube roots, we cube the expression for y:

$$y^3 = (a+b)^3 = a^3+b^3+3ab(a+b)$$

Let's calculate the terms a^3, b^3, and ab:

$$a^3 = \left(\sqrt[3]{x+\sqrt{x^2+1}}\right)^3 = x+\sqrt{x^2+1}$$

$$b^3 = \left(\sqrt[3]{x-\sqrt{x^2+1}}\right)^3 = x-\sqrt{x^2+1}$$

Summing these gives:

$$a^3+b^3 = (x+\sqrt{x^2+1}) + (x-\sqrt{x^2+1}) = 2x$$

Next, we find the product ab:

$$\begin{aligned} ab &= \sqrt[3]{(x+\sqrt{x^2+1})(x-\sqrt{x^2+1})} \\ &= \sqrt[3]{x^2-(x^2+1)} \\ &= \sqrt[3]{-1} = -1 \end{aligned}$$

Now substitute these results back into the expression for y^3. Note that $(a+b)$ is just y.

$$\begin{aligned} y^3 &= (a^3+b^3) + 3(ab)(a+b) \\ y^3 &= 2x + 3(-1)y \\ y^3 &= 2x - 3y \end{aligned}$$

We have found a direct relationship between y and x. To find the inverse, we swap x and y:

$$x^3 = 2y - 3x$$

Finally, we solve for y:

$$\begin{aligned} x^3 + 3x &= 2y \\ y &= \frac{x^3+3x}{2} \end{aligned}$$

Therefore, the inverse function is:

$$f^{-1}(x) = \frac{x^3+3x}{2}$$

***4.** Consider the function $f(x) = x^3 - 3x$.

(a) By restricting the domain to $x \in [1,2]$, the function is strictly increasing. Find the inverse function $f^{-1}(x)$ for this restricted domain.

Hint: $\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$.

Solution:
Let $y = x^3 - 3x$. The hint suggests a trigonometric substitution. We can scale the identity $\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$ by multiplying by 2:

$$2\cos(3\theta) = 8\cos^3\theta - 6\cos\theta = (2\cos\theta)^3 - 3(2\cos\theta)$$

This has the same form as our function $y = x^3 - 3x$. This suggests we let $x = 2\cos\theta$.
We must verify that this substitution covers the entire domain $x \in [1,2]$. For $x = 1$, we have $1 = 2\cos\theta \implies \cos\theta = 1/2 \implies \theta = \pi/3$. For $x = 2$, we have $2 = 2\cos\theta \implies \cos\theta = 1 \implies \theta = 0$. As x increases from 1 to 2, $\cos\theta$ increases from $1/2$ to 1, which means θ must decrease from $\pi/3$ to 0. So the domain for θ is $[0, \pi/3]$.

Substituting $x = 2\cos\theta$ into the function gives:

$$y = (2\cos\theta)^3 - 3(2\cos\theta) = 2\cos(3\theta)$$

We now have a system relating x, y, and θ:

$$\begin{aligned} x &= 2\cos\theta \\ y &= 2\cos(3\theta) \end{aligned}$$

To find the inverse function, we need to express x in terms of y. From the second equation, we solve for θ:

$$\begin{aligned} \frac{y}{2} &= \cos(3\theta) \\ 3\theta &= \arccos\left(\frac{y}{2}\right) \\ \theta &= \frac{1}{3}\arccos\left(\frac{y}{2}\right) \end{aligned}$$

Now substitute this expression for θ back into the first equation:

$$x = 2\cos\left(\frac{1}{3}\arccos\left(\frac{y}{2}\right)\right)$$

By swapping x and y, we get the inverse function:

$$f^{-1}(x) = 2\cos\left(\frac{1}{3}\arccos\left(\frac{x}{2}\right)\right)$$

(b) Now, consider the same function but restrict its domain to $x \in [2, \infty)$, where it is also strictly increasing. Find the inverse function $f^{-1}(x)$ for this new restricted domain.

Hint: A similar identity exists for the hyperbolic cosine function, $\cosh(t) = \frac{e^t+e^{-t}}{2}$. *The identity is* $\cosh(3t) = 4\cosh^3 t - 3\cosh t$.

Solution:

Following the same logic as part (a), we scale the hyperbolic identity:

$$2\cosh(3t) = 8\cosh^3 t - 6\cosh t = (2\cosh t)^3 - 3(2\cosh t)$$

This suggests the substitution $x = 2\cosh(t)$. Let's check the domain. The range of $\cosh(t)$ for $t \in [0, \infty)$ is $[1, \infty)$. Therefore, the range of $x = 2\cosh(t)$ for $t \in [0, \infty)$ is $[2, \infty)$, which perfectly matches the restricted domain.

Let $y = x^3 - 3x$. Substituting $x = 2\cosh(t)$:

$$y = (2\cosh t)^3 - 3(2\cosh t) = 2\cosh(3t)$$

From $y = 2\cosh(3t)$, we solve for t:

$$\begin{aligned} \frac{y}{2} &= \cosh(3t) \\ 3t &= \operatorname{arccosh}\left(\frac{y}{2}\right) \\ t &= \frac{1}{3}\operatorname{arccosh}\left(\frac{y}{2}\right) \end{aligned}$$

Now substitute this into $x = 2\cosh(t)$:

$$x = 2\cosh\left(\frac{1}{3}\text{arccosh}\left(\frac{y}{2}\right)\right)$$

Swapping x and y gives the inverse function for this domain:

$$f^{-1}(x) = 2\cosh\left(\frac{1}{3}\text{arccosh}\left(\frac{x}{2}\right)\right)$$

2. Limits

5. $\lim_{x\to\infty}\left(\sqrt{x+3\sqrt{x}} - \sqrt{x-\sqrt{x}}\right)$ Solution:

This limit is of the indeterminate form $\infty - \infty$. To resolve this, we multiply by the conjugate:

$$\begin{aligned}
&\lim_{x\to\infty}\left(\sqrt{x+3\sqrt{x}} - \sqrt{x-\sqrt{x}}\right) \cdot \frac{\sqrt{x+3\sqrt{x}} + \sqrt{x-\sqrt{x}}}{\sqrt{x+3\sqrt{x}} + \sqrt{x-\sqrt{x}}} \\
&= \lim_{x\to\infty} \frac{(x+3\sqrt{x}) - (x-\sqrt{x})}{\sqrt{x+3\sqrt{x}} + \sqrt{x-\sqrt{x}}} \\
&= \lim_{x\to\infty} \frac{4\sqrt{x}}{\sqrt{x+3\sqrt{x}} + \sqrt{x-\sqrt{x}}}
\end{aligned}$$

Now, we can factor out $\sqrt{x}$ from the denominator:

$$\begin{aligned}
&= \lim_{x\to\infty} \frac{4\sqrt{x}}{\sqrt{x(1+3/\sqrt{x})} + \sqrt{x(1-1/\sqrt{x})}} \\
&= \lim_{x\to\infty} \frac{4\sqrt{x}}{\sqrt{x}\left(\sqrt{1+3/\sqrt{x}} + \sqrt{1-1/\sqrt{x}}\right)} \\
&= \lim_{x\to\infty} \frac{4}{\sqrt{1+\frac{3}{\sqrt{x}}} + \sqrt{1-\frac{1}{\sqrt{x}}}}
\end{aligned}$$

As $x \to \infty$, the terms $3/\sqrt{x}$ and $1/\sqrt{x}$ go to 0.

$$= \frac{4}{\sqrt{1+0} + \sqrt{1-0}} = \frac{4}{1+1} = 2$$

6. $\lim_{x\to\infty} \frac{x^2+5}{2x+1} \sin\left(\frac{3}{x}\right)$ Solution:

We can rearrange the expression to make use of the fundamental trigonometric limit $\lim_{u\to 0} \frac{\sin u}{u} = 1$. Let $u = 3/x$. As $x \to \infty$, $u \to 0$.

$$\begin{aligned}
&\lim_{x\to\infty} \frac{x^2+5}{2x+1} \sin\left(\frac{3}{x}\right) \\
&= \lim_{x\to\infty} \frac{x^2+5}{2x+1} \cdot \frac{\sin(3/x)}{3/x} \cdot \frac{3}{x} \\
&= \lim_{x\to\infty} \left(\frac{3(x^2+5)}{x(2x+1)} \cdot \frac{\sin(3/x)}{3/x} \right)
\end{aligned}$$

We can split this into the product of two limits:

$$= \left(\lim_{x\to\infty} \frac{3x^2+15}{2x^2+x} \right) \cdot \left(\lim_{x\to\infty} \frac{\sin(3/x)}{3/x} \right)$$

The first limit is a rational function where the degrees of the numerator and denominator are equal, so the limit is the ratio of the leading coefficients:

$$\lim_{x\to\infty} \frac{3x^2+15}{2x^2+x} = \frac{3}{2}$$

The second limit is of the form $\lim_{u\to 0} \frac{\sin u}{u}$, which is 1.

$$\left(\frac{3}{2}\right) \cdot (1) = \frac{3}{2}$$

7. $\lim_{x\to 0} x^{-3} \left[\left(\frac{2+\cos x}{3} \right)^x - 1 \right]$ Solution:

We can rewrite the limit as:

$$\lim_{x\to 0} \frac{\left(\frac{2+\cos x}{3}\right)^x - 1}{x^3}$$

This is an indeterminate form $0/0$. Let's analyze the numerator. It is of the form $e^{f(x)} - 1$, and we know $\lim_{u\to 0} \frac{e^u-1}{u} = 1$. This suggests the limit is equivalent to the limit of its exponent divided by x^3. The exponent is $\ln\left[\left(\frac{2+\cos x}{3}\right)^x\right] = x \ln\left(\frac{2+\cos x}{3}\right)$. So the original limit is equivalent to:

$$\lim_{x\to 0} \frac{x \ln\left(\frac{2+\cos x}{3}\right)}{x^3} = \lim_{x\to 0} \frac{\ln\left(\frac{2+\cos x}{3}\right)}{x^2}$$

This is still a $0/0$ form, so we can apply L'Hôpital's Rule:

$$\begin{aligned}
&\lim_{x\to 0} \frac{\frac{d}{dx}\left[\ln\left(\frac{2+\cos x}{3}\right)\right]}{\frac{d}{dx}[x^2]} \\
&= \lim_{x\to 0} \frac{\frac{3}{2+\cos x} \cdot \left(\frac{-\sin x}{3}\right)}{2x} \\
&= \lim_{x\to 0} \frac{-\sin x}{2x(2+\cos x)}
\end{aligned}$$

We can separate this into a product of known limits:

$$= \lim_{x\to 0} \left(\frac{\sin x}{x}\right) \cdot \frac{-1}{2(2+\cos x)} = (1) \cdot \frac{-1}{2(2+1)} = -\frac{1}{6}$$

8. $\lim_{x\to\infty} (\ln x)^{\frac{1}{x}}$ Solution:

This is an indeterminate form ∞^0. Let $L = \lim_{x\to\infty}(\ln x)^{1/x}$. We can evaluate the limit of its logarithm:

$$\ln L = \lim_{x\to\infty} \ln\left((\ln x)^{1/x}\right) = \lim_{x\to\infty} \frac{\ln(\ln x)}{x}$$

This is an ∞/∞ form, so we can use L'Hôpital's Rule:

$$\begin{aligned} \ln L &= \lim_{x\to\infty} \frac{\frac{d}{dx}[\ln(\ln x)]}{\frac{d}{dx}[x]} \\ &= \lim_{x\to\infty} \frac{\frac{1}{\ln x}\cdot\frac{1}{x}}{1} \\ &= \lim_{x\to\infty} \frac{1}{x\ln x} = 0 \end{aligned}$$

Since $\ln L = 0$, we have $L = e^0 = 1$.

9. $\lim_{x\to 0} \frac{\tan x - \sin x}{x^3}$ Solution:

We can rewrite $\tan x$ as $\sin x/\cos x$:

$$\begin{aligned} &\lim_{x\to 0} \frac{\frac{\sin x}{\cos x} - \sin x}{x^3} \\ =&\lim_{x\to 0} \frac{\sin x\left(\frac{1}{\cos x} - 1\right)}{x^3} \\ =&\lim_{x\to 0} \frac{\sin x(1-\cos x)}{x^3\cos x} \end{aligned}$$

Now, we can split this into a product of well-known limits:

$$= \left(\lim_{x\to 0}\frac{\sin x}{x}\right) \cdot \left(\lim_{x\to 0}\frac{1-\cos x}{x^2}\right) \cdot \left(\lim_{x\to 0}\frac{1}{\cos x}\right)$$

We know that $\lim_{x\to 0}\frac{\sin x}{x} = 1$ and $\lim_{x\to 0}\frac{1-\cos x}{x^2} = \frac{1}{2}$. The third limit is $\frac{1}{\cos 0} = 1$.

$$= (1) \cdot \left(\frac{1}{2}\right) \cdot (1) = \frac{1}{2}$$

10. $\lim_{x\to 0^+} \left[\frac{(1+x)^{\frac{1}{x}}}{e}\right]^{\frac{1}{x}}$ Solution:

Let L be the limit. We analyze its logarithm:

$$\begin{aligned}
\ln L &= \lim_{x\to 0^+} \ln\left(\left[\frac{(1+x)^{1/x}}{e}\right]^{1/x}\right) \\
&= \lim_{x\to 0^+} \frac{1}{x} \ln\left[\frac{(1+x)^{1/x}}{e}\right] \\
&= \lim_{x\to 0^+} \frac{1}{x} \left[\ln\left((1+x)^{1/x}\right) - \ln(e)\right] \\
&= \lim_{x\to 0^+} \frac{1}{x} \left[\frac{1}{x}\ln(1+x) - 1\right] \\
&= \lim_{x\to 0^+} \frac{\ln(1+x) - x}{x^2}
\end{aligned}$$

This is a 0/0 indeterminate form, so we can apply L'Hôpital's Rule:

$$\begin{aligned}
\ln L &= \lim_{x\to 0^+} \frac{\frac{d}{dx}[\ln(1+x) - x]}{\frac{d}{dx}[x^2]} \\
&= \lim_{x\to 0^+} \frac{\frac{1}{1+x} - 1}{2x} \\
&= \lim_{x\to 0^+} \frac{\frac{1-(1+x)}{1+x}}{2x} \\
&= \lim_{x\to 0^+} \frac{-x}{2x(1+x)} = \lim_{x\to 0^+} \frac{-1}{2(1+x)} = -\frac{1}{2}
\end{aligned}$$

Since $\ln L = -1/2$, the limit is $L = e^{-1/2} = \frac{1}{\sqrt{e}}$.

11. $\lim_{x\to\alpha} \frac{\ln x - \ln\alpha}{x-\alpha}$, $(\alpha > 0)$ Solution:

This limit is the definition of the derivative of the function $f(x) = \ln x$ at the point $x = \alpha$.

$$f'(\alpha) = \lim_{x\to\alpha} \frac{f(x) - f(\alpha)}{x-\alpha} = \lim_{x\to\alpha} \frac{\ln x - \ln\alpha}{x-\alpha}$$

To find the value, we first compute the derivative of $f(x)$:

$$f'(x) = \frac{d}{dx}(\ln x) = \frac{1}{x}$$

Then we evaluate this derivative at $x = \alpha$:

$$f'(\alpha) = \frac{1}{\alpha}$$

12. $\lim_{x\to 0^+} \dfrac{1-\sqrt{\cos x}}{x(1-\cos\sqrt{x})}$ Solution:

We resolve the square root in the numerator by multiplying by its conjugate, $1 + \sqrt{\cos x}$:

$$\lim_{x\to 0^+} \frac{1-\sqrt{\cos x}}{x(1-\cos\sqrt{x})} \cdot \frac{1+\sqrt{\cos x}}{1+\sqrt{\cos x}} = \lim_{x\to 0^+} \frac{1-\cos x}{x(1-\cos\sqrt{x})(1+\sqrt{\cos x})}$$

Now we can regroup the terms to form standard limits:

$$= \lim_{x\to 0^+} \left(\frac{1-\cos x}{x^2}\right) \cdot \frac{x}{(1-\cos\sqrt{x})} \cdot \frac{1}{1+\sqrt{\cos x}}$$

The first term is a known limit: $\lim_{x\to 0^+} \frac{1-\cos x}{x^2} = \frac{1}{2}$. The third term is easy to evaluate: $\lim_{x\to 0^+} \frac{1}{1+\sqrt{\cos x}} = \frac{1}{1+\sqrt{1}} = \frac{1}{2}$. For the middle term, let $u = \sqrt{x}$. As $x \to 0^+$, $u \to 0^+$. The term is $\frac{(\sqrt{x})^2}{1-\cos\sqrt{x}} = \frac{u^2}{1-\cos u}$.

$$\lim_{u\to 0^+} \frac{u^2}{1-\cos u} = \frac{1}{\lim_{u\to 0^+} \frac{1-\cos u}{u^2}} = \frac{1}{1/2} = 2$$

Combining all the parts:

$$\left(\frac{1}{2}\right) \cdot (2) \cdot \left(\frac{1}{2}\right) = \frac{1}{2}$$

13. $\lim_{n\to\infty} n(\sqrt[n]{x} - 1), \quad (x > 0)$ Solution:

Let's make a substitution to transform this into a more familiar limit form. Let $h = 1/n$. As $n \to \infty$, $h \to 0$.

$$\lim_{n\to\infty} n(\sqrt[n]{x} - 1) = \lim_{h\to 0} \frac{1}{h}(x^h - 1) = \lim_{h\to 0} \frac{x^h - 1}{h}$$

This is the definition of the derivative of the function $f(t) = x^t$ at the point $t = 0$.

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{x^h - x^0}{h} = \lim_{h\to 0} \frac{x^h - 1}{h}$$

The derivative of $f(t) = x^t$ is $f'(t) = x^t \ln x$. Evaluating at $t = 0$:

$$f'(0) = x^0 \ln x = (1) \ln x = \ln x$$

14. Prove that $\lim_{n\to\infty} \sum_{i=0}^{n} \dfrac{n}{n^2 + i\pi} = 1$. Solution:

Let $S_n = \sum_{i=0}^{n} \frac{n}{n^2+i\pi}$. We have $n + 1$ terms in the sum. We can use the Squeeze Theorem by finding lower and upper bounds for the sum. The terms in the sum decrease as i increases. The largest term occurs at $i = 0$: $\frac{n}{n^2}$. The smallest term occurs at $i = n$: $\frac{n}{n^2+n\pi}$.

For the upper bound, we replace every term with the largest term:

$$S_n \leq \sum_{i=0}^{n} \frac{n}{n^2} = (n+1) \cdot \frac{n}{n^2} = \frac{n^2+n}{n^2} = 1 + \frac{1}{n}$$

The limit of the upper bound is $\lim_{n\to\infty}(1 + \frac{1}{n}) = 1$.

For the lower bound, we replace every term with the smallest term:

$$S_n \geq \sum_{i=0}^{n} \frac{n}{n^2+n\pi} = (n+1) \cdot \frac{n}{n^2+n\pi} = \frac{n^2+n}{n^2+n\pi}$$

The limit of the lower bound is $\lim_{n\to\infty} \frac{n^2+n}{n^2+n\pi} = \lim_{n\to\infty} \frac{1+1/n}{1+\pi/n} = \frac{1}{1} = 1$.

Since $1 \leq \lim_{n\to\infty} S_n \leq 1$, by the Squeeze Theorem, the limit must be 1.

15. Prove that $\displaystyle\lim_{n\to\infty} \sum_{i=0}^{n} \ln\left(1 + \frac{i}{n^2}\right) = \frac{1}{2}$. *Hint:* $x - \frac{x^2}{2} < \ln(1+x) < x$ for $x > 0$.

Solution:

Let $S_n = \sum_{i=0}^{n} \ln\left(1 + \frac{i}{n^2}\right)$. We apply the Squeeze Theorem using the given inequalities. For the upper bound, we use $\ln(1+x) < x$:

$$S_n < \sum_{i=0}^{n} \frac{i}{n^2} = \frac{1}{n^2} \sum_{i=0}^{n} i = \frac{1}{n^2}\left(\frac{n(n+1)}{2}\right) = \frac{n+1}{2n} = \frac{1}{2} + \frac{1}{2n}$$

The limit of the upper bound is $\lim_{n\to\infty}(\frac{1}{2} + \frac{1}{2n}) = \frac{1}{2}$.

For the lower bound, we use $\ln(1+x) > x - \frac{x^2}{2}$:

$$S_n > \sum_{i=0}^{n} \left[\frac{i}{n^2} - \frac{1}{2}\left(\frac{i}{n^2}\right)^2\right] = \sum_{i=0}^{n} \frac{i}{n^2} - \frac{1}{2n^4} \sum_{i=0}^{n} i^2$$

From the upper bound calculation, we know $\sum \frac{i}{n^2} = \frac{n+1}{2n}$. The second term uses the formula for the sum of squares, $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$:

$$\frac{1}{2n^4} \sum_{i=0}^{n} i^2 = \frac{1}{2n^4} \frac{n(n+1)(2n+1)}{6} = \frac{2n^3+3n^2+n}{12n^4}$$

The limit of the lower bound is:

$$\lim_{n\to\infty} \left(\frac{n+1}{2n} - \frac{2n^3+3n^2+n}{12n^4}\right) = \frac{1}{2} - 0 = \frac{1}{2}$$

Since $\frac{1}{2} \leq \lim_{n\to\infty} S_n \leq \frac{1}{2}$, the limit must be $\frac{1}{2}$.

16. (MARK: Use of L'Hôpital's rule is extremely tedious here) Calculate the limit:

$$\lim_{x\to 0} \frac{e^{x^2} - \cos(x) - \frac{3}{2}x^2}{x^4}$$

Solution:

This is a 0/0 indeterminate form. We must apply L'Hôpital's Rule four times. Let $N(x) = e^{x^2} - \cos(x) - \frac{3}{2}x^2$ and $D(x) = x^4$.

$$\begin{aligned}
\lim_{x\to 0} \frac{N(x)}{D(x)} &= \lim_{x\to 0} \frac{2xe^{x^2} + \sin(x) - 3x}{4x^3} \quad (0/0) \\
&= \lim_{x\to 0} \frac{2e^{x^2} + 4x^2e^{x^2} + \cos(x) - 3}{12x^2} \quad (0/0) \\
&= \lim_{x\to 0} \frac{4xe^{x^2} + 8xe^{x^2} + 8x^3e^{x^2} - \sin(x)}{24x} \\
&= \lim_{x\to 0} \frac{12xe^{x^2} + 8x^3e^{x^2} - \sin(x)}{24x} \quad (0/0) \\
&= \lim_{x\to 0} \frac{12e^{x^2} + 24x^2e^{x^2} + 24x^2e^{x^2} + 16x^4e^{x^2} - \cos(x)}{24} \\
&= \frac{12e^0 + 0 + 0 + 0 - \cos(0)}{24} = \frac{12-1}{24} = \frac{11}{24}
\end{aligned}$$

Note: This problem is designed to be solved efficiently using Taylor expansions. The repeated application of L'Hôpital's Rule, while correct, is very labor-intensive.

17. Let $a, b > 0$. Calculate the limit:

$$\lim_{x\to 0} \left(\frac{a^x + b^x}{2} \right)^{\frac{1}{x}}$$

Solution:

This is a 1^∞ indeterminate form. Let L be the limit. We evaluate its logarithm:

$$\ln L = \lim_{x\to 0} \ln \left[\left(\frac{a^x + b^x}{2} \right)^{\frac{1}{x}} \right] = \lim_{x\to 0} \frac{1}{x} \ln \left(\frac{a^x + b^x}{2} \right)$$

This is a 0/0 form, so we use L'Hôpital's Rule:

$$\begin{aligned}
\ln L &= \lim_{x\to 0} \frac{\frac{d}{dx}\left[\ln\left(\frac{a^x+b^x}{2}\right)\right]}{\frac{d}{dx}[x]} \\
&= \lim_{x\to 0} \frac{\frac{2}{a^x+b^x} \cdot \frac{a^x \ln a + b^x \ln b}{2}}{1} \\
&= \frac{\frac{2}{a^0+b^0} \cdot \frac{a^0 \ln a + b^0 \ln b}{2}}{1} \\
&= \frac{\frac{2}{1+1} \cdot \frac{\ln a + \ln b}{2}}{1} = \frac{\ln a + \ln b}{2} = \frac{\ln(ab)}{2} = \ln(\sqrt{ab})
\end{aligned}$$

Since $\ln L = \ln(\sqrt{ab})$, the limit is $L = \sqrt{ab}$.

18. Evaluate the following limit related to Stirling's approximation:

$$\lim_{n\to\infty} \frac{\sqrt[n]{n!}}{n}$$

Hint: You may use Stirling's approximation, $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$. Solution:

We will provide two methods to solve this.

Method 1: Using Stirling's Approximation

Let $L = \lim_{n\to\infty} \frac{(n!)^{1/n}}{n}$. Consider its logarithm:

$$\ln L = \lim_{n\to\infty} \ln\left(\frac{(n!)^{1/n}}{n}\right) = \lim_{n\to\infty}\left[\frac{1}{n}\ln(n!) - \ln n\right]$$

Using the approximation $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$, we get:

$$\begin{aligned}
\ln(n!) &\approx \ln\left(\sqrt{2\pi n}\left(\frac{n}{e}\right)^n\right) \\
&= \ln(\sqrt{2\pi n}) + \ln\left(\left(\frac{n}{e}\right)^n\right) \\
&= \frac{1}{2}\ln(2\pi n) + n\ln\left(\frac{n}{e}\right) \\
&= \frac{1}{2}\ln(2\pi) + \frac{1}{2}\ln(n) + n(\ln n - \ln e) \\
&= \frac{1}{2}\ln(2\pi) + \frac{1}{2}\ln(n) + n\ln n - n
\end{aligned}$$

Substitute this back into the expression for $\ln L$:

$$\begin{aligned}
\ln L &= \lim_{n\to\infty}\left[\frac{1}{n}\left(\frac{\ln(2\pi)}{2} + \frac{\ln n}{2} + n\ln n - n\right) - \ln n\right] \\
&= \lim_{n\to\infty}\left[\frac{\ln(2\pi)}{2n} + \frac{\ln n}{2n} + \ln n - 1 - \ln n\right] \\
&= \lim_{n\to\infty}\left[\frac{\ln(2\pi)}{2n} + \frac{\ln n}{2n} - 1\right]
\end{aligned}$$

As $n \to \infty$, both $\frac{\ln(2\pi)}{2n}$ and $\frac{\ln n}{2n}$ go to 0.

$$\ln L = 0 + 0 - 1 = -1$$

Since $\ln L = -1$, the limit is $L = e^{-1} = \frac{1}{e}$.

Method 2: Using Riemann Sums

Let $L = \lim_{n\to\infty} \frac{(n!)^{1/n}}{n}$. The logarithm is:

$$
\begin{aligned}
\ln L &= \lim_{n\to\infty} \left[\frac{1}{n}\ln(n!) - \ln n\right] \\
&= \lim_{n\to\infty} \left[\frac{1}{n}\ln(1\cdot 2\cdot \cdots \cdot n) - \ln n\right] \\
&= \lim_{n\to\infty} \left[\frac{1}{n}\sum_{k=1}^{n}\ln k - \frac{1}{n}\sum_{k=1}^{n}\ln n\right] \\
&= \lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n}(\ln k - \ln n) = \lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n}\ln\left(\frac{k}{n}\right)
\end{aligned}
$$

This is the definition of a Riemann sum for the function $f(x) = \ln x$ on the interval $[0, 1]$. Here, $\Delta x = 1/n$ and $x_k = k/n$.

$$
\ln L = \int_0^1 \ln(x)\,dx
$$

We use integration by parts ($\int udv = uv - \int vdu$) with $u = \ln x, dv = dx$:

$$
\begin{aligned}
\int_0^1 \ln(x)\,dx &= [x\ln x - x]_0^1 \\
&= (1\ln 1 - 1) - \lim_{a\to 0^+}(a\ln a - a) \\
&= (-1) - (0 - 0) = -1
\end{aligned}
$$

(Note: $\lim_{a\to 0^+} a\ln a = 0$ can be shown with L'Hôpital's Rule). Since $\ln L = -1$, the limit is $L = e^{-1} = \frac{1}{e}$.

Historical Note: *Stirling's approximation is a powerful formula for approximating the factorial function for large n. It was first discovered by the Scottish mathematician James Stirling and independently by Abraham de Moivre. The connection between the discrete sum $\ln(n!)$ and the continuous integral $\int \ln x dx$ shown in Method 2 is a key idea in its derivation.*

19. (MARK: Taylor series is the only feasible method here) Calculate the following second-order limit involving the number e:

$$
\lim_{x\to 0} \frac{(1+x)^{1/x} - e + \frac{e}{2}x}{x^2}
$$

Solution:

Let $f(x) = (1+x)^{1/x}$. This limit requires a high-order approximation of $f(x)$ around $x = 0$. This is most easily achieved with a Taylor expansion. We start with the logarithm of $f(x)$:

$$
\ln(f(x)) = \frac{\ln(1+x)}{x} = \frac{1}{x}\left(x - \frac{x^2}{2} + \frac{x^3}{3} - O(x^4)\right) = 1 - \frac{x}{2} + \frac{x^2}{3} - O(x^3)
$$

Now, we exponentiate this result:

$$f(x) = e^{\ln(f(x))} = e^{1-\frac{x}{2}+\frac{x^2}{3}-\dots} = e \cdot e^{-\frac{x}{2}+\frac{x^2}{3}-\dots}$$

Using the expansion $e^u = 1 + u + \frac{u^2}{2!} + O(u^3)$ with $u = -\frac{x}{2} + \frac{x^2}{3}$:

$$\begin{aligned}
f(x) &= e\left[1 + \left(-\frac{x}{2} + \frac{x^2}{3}\right) + \frac{1}{2}\left(-\frac{x}{2} + \frac{x^2}{3}\right)^2 + O(x^3)\right] \\
&= e\left[1 - \frac{x}{2} + \frac{x^2}{3} + \frac{1}{2}\left(\frac{x^2}{4} - \dots\right) + O(x^3)\right] \\
&= e\left[1 - \frac{x}{2} + \left(\frac{1}{3} + \frac{1}{8}\right)x^2 + O(x^3)\right] \\
&= e - \frac{e}{2}x + \frac{11e}{24}x^2 + O(x^3)
\end{aligned}$$

Now substitute this expansion into the numerator of the limit:

$$\begin{aligned}
&\left(e - \frac{e}{2}x + \frac{11e}{24}x^2 + O(x^3)\right) - e + \frac{e}{2}x \\
=&\frac{11e}{24}x^2 + O(x^3)
\end{aligned}$$

The limit becomes:

$$\lim_{x\to 0} \frac{\frac{11e}{24}x^2 + O(x^3)}{x^2} = \lim_{x\to 0}\left(\frac{11e}{24} + O(x)\right) = \frac{11e}{24}$$

20. Evaluate the following limits:

$$\lim_{x\to 0^+} [\cos(2x)]^{1/x^2}$$

Solution:

This is a 1^∞ indeterminate form. Let L be the limit. We evaluate its logarithm:

$$\ln L = \lim_{x\to 0^+} \ln\left([\cos(2x)]^{1/x^2}\right) = \lim_{x\to 0^+} \frac{\ln(\cos(2x))}{x^2}$$

This is a $0/0$ form, so we use L'Hôpital's Rule:

$$\begin{aligned}
\ln L &= \lim_{x\to 0^+} \frac{\frac{d}{dx}[\ln(\cos(2x))]}{\frac{d}{dx}[x^2]} \\
&= \lim_{x\to 0^+} \frac{\frac{1}{\cos(2x)} \cdot (-2\sin(2x))}{2x} \\
&= \lim_{x\to 0^+} \frac{-\sin(2x)}{x\cos(2x)} \\
&= \lim_{x\to 0^+} \left(\frac{\sin(2x)}{2x} \cdot \frac{-2}{\cos(2x)}\right)
\end{aligned}$$

The first part is a standard limit that goes to 1. The second part goes to $-2/\cos(0) = -2$.

$$\ln L = (1) \cdot (-2) = -2$$

Thus, the limit is $L = e^{-2}$.

21. Evaluate the following limits:

$$\lim_{t\to\infty} \left[t \ln \left(1 + \frac{3}{t} \right) \right]$$

Solution:

This is an $\infty \cdot 0$ indeterminate form. We can rewrite it as a fraction. Let $h = 1/t$. As $t \to \infty$, $h \to 0$.

$$\lim_{t\to\infty} t \ln \left(1 + \frac{3}{t} \right) = \lim_{h\to 0} \frac{1}{h} \ln(1 + 3h) = \lim_{h\to 0} \frac{\ln(1+3h)}{h}$$

This is a 0/0 form. Applying L'Hôpital's Rule:

$$= \lim_{h\to 0} \frac{\frac{d}{dh}[\ln(1+3h)]}{\frac{d}{dh}[h]} = \lim_{h\to 0} \frac{\frac{1}{1+3h} \cdot 3}{1} = \frac{3}{1+0} = 3$$

Alternatively, this is the definition of the derivative of $f(x) = \ln(1 + 3x)$ at $x = 0$. $f'(x) = \frac{3}{1+3x}$, so $f'(0) = 3$.

3. Derivatives and Continuity

22. The Folium of Descartes is given by the equation $x^3 + y^3 = 6xy$. Find the value of the second derivative, $\frac{d^2y}{dx^2}$, at the point $(3, 3)$. Solution:

First, we find the first derivative, $y' = \frac{dy}{dx}$, using implicit differentiation.

$$\frac{d}{dx}(x^3 + y^3) = \frac{d}{dx}(6xy)$$

$$3x^2 + 3y^2y' = 6y + 6xy'$$

Now, evaluate at the point $(3, 3)$ to find the slope at that point:

$$\begin{aligned} 3(3)^2 + 3(3)^2y' &= 6(3) + 6(3)y' \\ 27 + 27y' &= 18 + 18y' \\ 9y' &= -9 \\ y' &= -1 \end{aligned}$$

Next, we differentiate the expression $3x^2 + 3y^2y' = 6y + 6xy'$ again with respect to x:

$$\frac{d}{dx}(3x^2 + 3y^2y') = \frac{d}{dx}(6y + 6xy')$$

Using the product rule on the terms $3y^2y'$ and $6xy'$:

$$6x + (6y \cdot y' \cdot y' + 3y^2 \cdot y'') = 6y' + (6y' + 6x \cdot y'')$$
$$6x + 6y(y')^2 + 3y^2y'' = 12y' + 6xy''$$

Now, we solve for y'':

$$\begin{aligned} 3y^2y'' - 6xy'' &= 12y' - 6x - 6y(y')^2 \\ y''(3y^2 - 6x) &= 12y' - 6x - 6y(y')^2 \end{aligned}$$

Finally, substitute the point $(3,3)$ and the value $y' = -1$ into this equation:

$$\begin{aligned} y''(3(3)^2 - 6(3)) &= 12(-1) - 6(3) - 6(3)(-1)^2 \\ y''(27 - 18) &= -12 - 18 - 18 \\ 9y'' &= -48 \\ y'' &= -\frac{48}{9} = -\frac{16}{3} \end{aligned}$$

Historical Note: *The Folium of Descartes (Latin for "leaf of Descartes") is a famous curve first proposed by René Descartes in 1638. Its exploration was significant in the development of analytic geometry and calculus.*

23. The Witch of Agnesi curve is given by the equation $y = \frac{8}{x^2+4}$. Find the coordinates of its inflection points. Solution:

Inflection points occur where the second derivative, y'', changes sign. We first find y' and y''.

$$y = 8(x^2 + 4)^{-1}$$
$$y' = -8(x^2 + 4)^{-2}(2x) = -16x(x^2 + 4)^{-2}$$

Now, find the second derivative using the product rule and chain rule:

$$\begin{aligned} y'' &= -16(x^2 + 4)^{-2} + (-16x)\left(-2(x^2 + 4)^{-3}(2x)\right) \\ &= -16(x^2 + 4)^{-2} + 64x^2(x^2 + 4)^{-3} \\ &= \frac{-16}{(x^2 + 4)^2} + \frac{64x^2}{(x^2 + 4)^3} \\ &= \frac{-16(x^2 + 4) + 64x^2}{(x^2 + 4)^3} \\ &= \frac{-16x^2 - 64 + 64x^2}{(x^2 + 4)^3} = \frac{48x^2 - 64}{(x^2 + 4)^3} \end{aligned}$$

Set $y'' = 0$ to find potential inflection points:

$$\begin{aligned} 48x^2 - 64 &= 0 \\ 48x^2 &= 64 \\ x^2 &= \frac{64}{48} = \frac{4}{3} \\ x &= \pm\frac{2}{\sqrt{3}} \end{aligned}$$

Now, find the corresponding y-coordinates:

$$y = \frac{8}{(\pm 2/\sqrt{3})^2 + 4} = \frac{8}{4/3 + 4} = \frac{8}{16/3} = 8 \cdot \frac{3}{16} = \frac{3}{2}$$

The inflection points are $\left(\frac{2}{\sqrt{3}}, \frac{3}{2}\right)$ and $\left(-\frac{2}{\sqrt{3}}, \frac{3}{2}\right)$.

Historical Note: *The Witch of Agnesi is a curve studied by Italian mathematician Maria Gaetana Agnesi in 1748. The name "witch" comes from a mistranslation of the Italian word for "versed sine curve," which sounded like the word for "witch's wife."*

24. A cycloid is given by the parametric equations:

$$x = r(\theta - \sin\theta), \quad y = r(1 - \cos\theta)$$

Find the second derivative, $\frac{d^2y}{dx^2}$, as a function of θ. Solution:

For parametric equations, the first derivative is $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$.

$$\frac{dy}{d\theta} = r\sin\theta$$

$$\frac{dx}{d\theta} = r(1 - \cos\theta)$$

$$\frac{dy}{dx} = \frac{r\sin\theta}{r(1 - \cos\theta)} = \frac{\sin\theta}{1 - \cos\theta}$$

Using half-angle identities $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$ and $1 - \cos\theta = 2\sin^2(\theta/2)$:

$$\frac{dy}{dx} = \frac{2\sin(\theta/2)\cos(\theta/2)}{2\sin^2(\theta/2)} = \frac{\cos(\theta/2)}{\sin(\theta/2)} = \cot(\theta/2)$$

The second derivative is $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{\frac{d}{d\theta}\left(\frac{dy}{dx}\right)}{dx/d\theta}$. First, find the derivative of y' with respect to θ:

$$\frac{d}{d\theta}(\cot(\theta/2)) = -\csc^2(\theta/2) \cdot \frac{1}{2}$$

Now, divide by $dx/d\theta$:

$$\begin{aligned}
\frac{d^2y}{dx^2} &= \frac{-\frac{1}{2}\csc^2(\theta/2)}{r(1 - \cos\theta)} \\
&= \frac{-\frac{1}{2}\csc^2(\theta/2)}{r(2\sin^2(\theta/2))} \\
&= \frac{-1}{4r\sin^2(\theta/2)\sin^2(\theta/2)} = \frac{-1}{4r\sin^4(\theta/2)}
\end{aligned}$$

Historical Note: *The cycloid is the curve traced by a point on a rolling circle. It is famous for being the solution to the brachistochrone problem (the path of fastest descent) and the tautochrone problem (an object released on it takes the same time to reach the bottom regardless of starting point).*

25. Given $\cos(x^2+2y)+xe^{y^2}=1$, find $\frac{dy}{dx}$ by implicit differentiation. Solution:

We differentiate both sides of the equation with respect to x. Remember to use the chain rule for terms involving y and the product rule for the term xe^{y^2}.

$$\frac{d}{dx}\left(\cos(x^2+2y)+xe^{y^2}\right)=\frac{d}{dx}(1)$$

Differentiating term by term:

$$\begin{aligned}\frac{d}{dx}(\cos(x^2+2y)) &= -\sin(x^2+2y)\cdot\frac{d}{dx}(x^2+2y)\\ &= -\sin(x^2+2y)\left(2x+2\frac{dy}{dx}\right)\end{aligned}$$

$$\begin{aligned}\frac{d}{dx}(xe^{y^2}) &= (1)e^{y^2}+x\cdot\frac{d}{dx}(e^{y^2})\\ &= e^{y^2}+x\left(e^{y^2}\cdot 2y\frac{dy}{dx}\right)\end{aligned}$$

Combining these results and setting the sum to zero (since the derivative of 1 is 0):

$$-\sin(x^2+2y)\left(2x+2\frac{dy}{dx}\right)+e^{y^2}+2xye^{y^2}\frac{dy}{dx}=0$$

Now, we expand and solve for $\frac{dy}{dx}$:

$$-2x\sin(x^2+2y)-2\sin(x^2+2y)\frac{dy}{dx}+e^{y^2}+2xye^{y^2}\frac{dy}{dx}=0$$

$$\frac{dy}{dx}\left(2xye^{y^2}-2\sin(x^2+2y)\right)=2x\sin(x^2+2y)-e^{y^2}$$

$$\frac{dy}{dx}=\frac{2x\sin(x^2+2y)-e^{y^2}}{2xye^{y^2}-2\sin(x^2+2y)}$$

26. Given $f:\mathbb{R}\to\mathbb{R}$, with

$$f(x)=\begin{cases}x^4\sin\frac{1}{x}, & x\neq 0\\ 0, & x=0\end{cases}$$

(a) Calculate $f'(x)$.

Solution:
For $x\neq 0$, we use the Product Rule and Chain Rule on $f(x)=x^4\sin\left(\frac{1}{x}\right)$:

$$\begin{aligned}f'(x) &= \frac{d}{dx}(x^4)\cdot\sin\left(\frac{1}{x}\right)+x^4\cdot\frac{d}{dx}\left(\sin\left(\frac{1}{x}\right)\right)\\ &= 4x^3\sin\left(\frac{1}{x}\right)+x^4\left(\cos\left(\frac{1}{x}\right)\cdot\left(-\frac{1}{x^2}\right)\right)\\ &= 4x^3\sin\left(\frac{1}{x}\right)-x^2\cos\left(\frac{1}{x}\right)\end{aligned}$$

For $x = 0$, we must use the limit definition of the derivative:

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{h^4 \sin\left(\frac{1}{h}\right) - 0}{h} = \lim_{h\to 0} h^3 \sin\left(\frac{1}{h}\right)$$

Since $-1 \leq \sin(1/h) \leq 1$, we have $-|h^3| \leq h^3 \sin(1/h) \leq |h^3|$. By the Squeeze Theorem, as $h \to 0$, the limit is 0. So, $f'(0) = 0$.

Combining these results, we get:

$$f'(x) = \begin{cases} 4x^3 \sin\frac{1}{x} - x^2 \cos\frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

(b) Calculate $f''(0)$.

Solution:

We use the limit definition for the second derivative at $x = 0$, applied to $f'(x)$:

$$\begin{aligned} f''(0) &= \lim_{h\to 0} \frac{f'(0+h) - f'(0)}{h} \\ &= \lim_{h\to 0} \frac{4h^3 \sin\left(\frac{1}{h}\right) - h^2 \cos\left(\frac{1}{h}\right) - 0}{h} \\ &= \lim_{h\to 0} \left(4h^2 \sin\left(\frac{1}{h}\right) - h \cos\left(\frac{1}{h}\right)\right) \end{aligned}$$

By the Squeeze Theorem, both $\lim_{h\to 0} 4h^2 \sin(1/h) = 0$ and $\lim_{h\to 0} h \cos(1/h) = 0$.

$$f''(0) = 0 - 0 = 0$$

Conceptual Note: *This function is an interesting example in calculus. Although $f(x)$ is twice differentiable everywhere (including at $x = 0$), its second derivative, $f''(x)$, is not continuous at $x = 0$. This illustrates that the existence of derivatives does not guarantee their continuity.*

27. Suppose that $f(x)$ is continuous near $x = 0$, satisfying $f(x) - \frac{1}{2}f(\frac{x}{2}) = x^2$. What is $f(x)$?

Solution:

We are given the functional equation $f(x) - \frac{1}{2}f(\frac{x}{2}) = x^2$. We can rearrange this to express $f(x)$ as:

$$f(x) = x^2 + \frac{1}{2}f\left(\frac{x}{2}\right)$$

We can apply this relation recursively:

$$\begin{aligned} f(x) &= x^2 + \frac{1}{2}\left[\left(\frac{x}{2}\right)^2 + \frac{1}{2}f\left(\frac{x}{4}\right)\right] \\ &= x^2 + \frac{x^2}{8} + \frac{1}{4}f\left(\frac{x}{4}\right) \\ &= x^2 + \frac{x^2}{8} + \frac{1}{4}\left[\left(\frac{x}{4}\right)^2 + \frac{1}{2}f\left(\frac{x}{8}\right)\right] \\ &= x^2 + \frac{x^2}{8} + \frac{x^2}{64} + \frac{1}{8}f\left(\frac{x}{8}\right) \end{aligned}$$

After n steps, we can see the pattern:

$$f(x) = x^2 \left(1 + \frac{1}{8} + \frac{1}{64} + \cdots + \frac{1}{8^{n-1}}\right) + \frac{1}{2^n} f\left(\frac{x}{2^n}\right)$$

As $n \to \infty$, the sum becomes an infinite geometric series with first term $a = 1$ and common ratio $r = 1/8$.

$$\sum_{k=0}^{\infty} \left(\frac{1}{8}\right)^k = \frac{1}{1 - 1/8} = \frac{8}{7}$$

For the second term, since f is continuous at 0, we first find $f(0)$ from the original equation: $f(0) - \frac{1}{2}f(0) = 0^2 \implies \frac{1}{2}f(0) = 0 \implies f(0) = 0$. As $n \to \infty$, $x/2^n \to 0$, so $\lim_{n\to\infty} f(x/2^n) = f(0) = 0$. This means the term $\frac{1}{2^n} f\left(\frac{x}{2^n}\right) \to 0$. Combining the results, we have:

$$f(x) = x^2 \left(\frac{8}{7}\right) + 0 = \frac{8}{7}x^2$$

Method Note: *This technique of repeated substitution is common for solving functional equations. By iterating the relationship, we transform the problem into finding the sum of an infinite series and evaluating a limit, which often simplifies the problem significantly.*

28. Suppose

$$f(x) = \begin{cases} \frac{x}{1+e^{\frac{1}{x}}} & (x \neq 0) \\ 0 & (x = 0) \end{cases}$$

Prove that $f(x)$ is continuous on $\mathbb{R}$.

Solution:

For any $x \neq 0$, the function is a composition and quotient of continuous functions (x, 1, e^u, $1/x$), and the denominator $1 + e^{1/x}$ is never zero. Thus, $f(x)$ is continuous for all $x \in \mathbb{R} \setminus \{0\}$.

The only point we need to check is $x = 0$. For continuity at $x = 0$, we must show that $\lim_{x\to 0} f(x) = f(0)$. Since $f(0) = 0$, we must verify that the limit is also zero. We check the left-hand and right-hand limits.

Right-hand limit ($x \to 0^+$): As $x \to 0^+$, the exponent $1/x \to +\infty$. Therefore, $e^{1/x} \to \infty$.

$$\lim_{x\to 0^+} \frac{x}{1 + e^{\frac{1}{x}}} \quad \to \quad \frac{0}{1+\infty} \quad \to \quad 0$$

Left-hand limit ($x \to 0^-$): As $x \to 0^-$, the exponent $1/x \to -\infty$. Therefore, $e^{1/x} \to 0$.

$$\lim_{x\to 0^-} \frac{x}{1 + e^{\frac{1}{x}}} = \frac{0}{1+0} = 0$$

Since $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x) = 0$, the overall limit is $\lim_{x\to 0} f(x) = 0$. Because this limit equals $f(0)$, the function is continuous at $x = 0$. Therefore, $f(x)$ is continuous on all of $\mathbb{R}$.

Conceptual Note: *This function exhibits a "jump" in its derivative at $x = 0$, even though the function itself is continuous. It's a good example of a function that is continuous but not differentiable at a point. The sharp change in behavior of $e^{1/x}$ around zero is key to this property.*

29. Suppose that f satisfies the equation

$$f(x+y) = f(x) + f(y) + x^2y + xy^2 \quad \text{for all } x, y \in \mathbb{R}$$

Suppose further that $\lim_{x\to 0} \frac{f(x)}{x} = 1$.

(a) Calculate $f'(0)$.

Solution:
The derivative at 0 is defined as $f'(0) = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h}$. First, we find $f(0)$ by setting $x = y = 0$ in the given equation:

$$f(0+0) = f(0) + f(0) + 0 \implies f(0) = 2f(0) \implies f(0) = 0$$

Now we can calculate the derivative:

$$f'(0) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{f(h)}{h}$$

We are given that this limit is 1. Therefore, $f'(0) = 1$.

(b) Calculate $f'(x)$.

Solution:
Using the limit definition for the derivative of $f(x)$:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

We use the functional equation to expand $f(x+h)$:

$$f(x+h) = f(x) + f(h) + x^2h + xh^2$$

Substitute this into the derivative definition:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{(f(x) + f(h) + x^2h + xh^2) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{f(h) + x^2h + xh^2}{h} \\ &= \lim_{h\to 0} \left(\frac{f(h)}{h} + x^2 + xh \right) \end{aligned}$$

Evaluating the limit of each term separately:

$$f'(x) = \left(\lim_{h\to 0} \frac{f(h)}{h} \right) + \left(\lim_{h\to 0} x^2 \right) + \left(\lim_{h\to 0} xh \right)$$

Using the given limit, we get:

$$f'(x) = 1 + x^2 + 0 = 1 + x^2$$

Problem Note: *This type of problem, combining a functional equation with a limit condition, is a classic way to determine the derivative of an unknown function. The structure of the functional equation often reveals the structure of the derivative.*

30. Suppose we know that $f(x)$ is continuous and differentiable on the interval $[-6,-1]$, that $f(-6) = -23$ and that $f'(x) \geq -4$. What is the smallest possible value for $f(-1)$?

Solution:

The problem provides conditions that are perfect for applying the Mean Value Theorem (MVT). The MVT states that for a function f continuous on $[a,b]$ and differentiable on (a,b), there exists a number $c \in (a,b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

In our case, $a = -6$ and $b = -1$. The theorem guarantees there is a $c \in (-6,-1)$ such that:

$$f'(c) = \frac{f(-1) - f(-6)}{-1 - (-6)} = \frac{f(-1) - (-23)}{5} = \frac{f(-1) + 23}{5}$$

We are given the condition that $f'(x) \geq -4$ for all x in the interval. This must also hold for the specific value c, so $f'(c) \geq -4$. We can set up an inequality:

$$\frac{f(-1) + 23}{5} \geq -4$$

Now, we solve for $f(-1)$:

$$\begin{aligned} f(-1) + 23 &\geq -20 \\ f(-1) &\geq -20 - 23 \\ f(-1) &\geq -43 \end{aligned}$$

The inequality shows that the value of $f(-1)$ must be greater than or equal to -43. Therefore, the smallest possible value for $f(-1)$ is **-43**.

Theorem Note: *The Mean Value Theorem is a cornerstone of calculus. It connects the average rate of change over an interval (the secant line) to the instantaneous rate of change at a specific point (the tangent line). This problem is a direct application of using the MVT to bound the value of a function.*